

Evaluate independent physical evidence, not the number of windows

Strong protocol if a third environment is available

Train specimens	TRAIN	TRAIN	reserved
Validation specimens	VALIDATE	VALIDATE	reserved
Test specimens	reserved	reserved	TEST
	E1	E2	E3 (unseen)
	Environment		

Current two-environment protocol

Train: E1 / train specimens
Validate: E1 / separate specimens
Test: E2 / held-out specimens

External motor cases

One healthy + one faulty motor
Device identity is confounded with health
Report case results, not population accuracy

Every partition keeps whole runs

No overlapping windows across splits
Train-only PCA / references / thresholds
Calibration P0 / P1 / P2 reported separately